

Broadband Photovoltaic Detectors Based on an Atomically Thin Heterostructure

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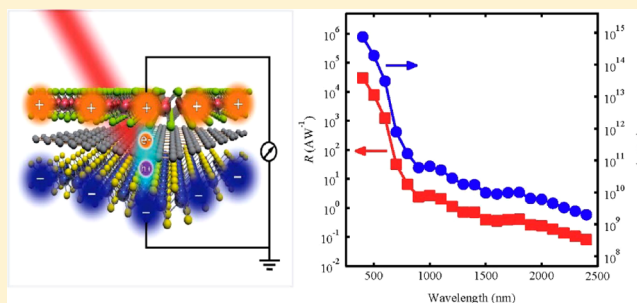
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Supporting Information

ABSTRACT: van der Waals junctions of two-dimensional materials with an atomically sharp interface open up unprecedented opportunities to design and study functional heterostructures. Semiconducting transition metal dichalcogenides have shown tremendous potential for future applications due to their unique electronic properties and strong light-matter interaction. However, many important optoelectronic applications, such as broadband photodetection, are severely hindered by their limited spectral range and reduced light absorption. Here, we present a p–g–n heterostructure formed by sandwiching graphene with a gapless band structure and wide absorption spectrum in an atomically thin p–n junction to overcome these major limitations. We have successfully demonstrated a MoS₂–graphene–WSe₂ heterostructure for broadband photodetection in the visible to short-wavelength infrared range at room temperature that exhibits competitive device performance, including a specific detectivity of up to 10¹¹ Jones in the near-infrared region. Our results pave the way toward the implementation of atomically thin heterostructures for broadband and sensitive optoelectronic applications.

KEYWORDS: 2D materials, transition-metal dichalcogenides, graphene, heterostructure, photodetection



Current photodetection technologies rely primarily on separate photoactive semiconducting materials with certain band gaps corresponding to distinct spectral ranges. The realization of broadband light detection is crucial in many important optoelectronic applications, such as sensing, imaging¹ and communication.^{2,3} Transition metal dichalcogenides (TMDs) have recently been demonstrated with remarkable optical^{4–19} and electronic^{20,21} properties, and they are expected to play an important role as an essential unit in future ultrascaled and integrated optoelectronic circuits.

However, broadband photodetection based on TMD materials remains a central challenge due to limited spectral bandwidth and reduced light absorption similar to conventional group IV and III–V semiconductors. In the past few years, intensive research efforts have been focused on creating van der Waals junctions by stacking selected two-dimensional (2D) materials. Due to atomically thin geometry and atomically sharp interface,^{22,23} many new emergent physical phenomenon are

expected in van der Waals heterostructures. Such approach offers tremendous opportunities for designing functional electronic^{16,24} and optoelectronic^{15,25–27} units with competitive performances.

In this work, we present a p–g–n heterostructure formed by sandwiching graphene (g) in an atomically thin p–n junction to realize broadband and high-sensitivity photovoltaic detectors. A schematic drawing is shown in Figure 1a. Graphene with a gapless band structure^{28–30} provides an effective absorption material over a wide spectral range. The built-in electric field formed at the depletion region of the p–n junction plays a crucial role in efficiently separating photoexcited electrons and holes and enables broadband photodetection with high

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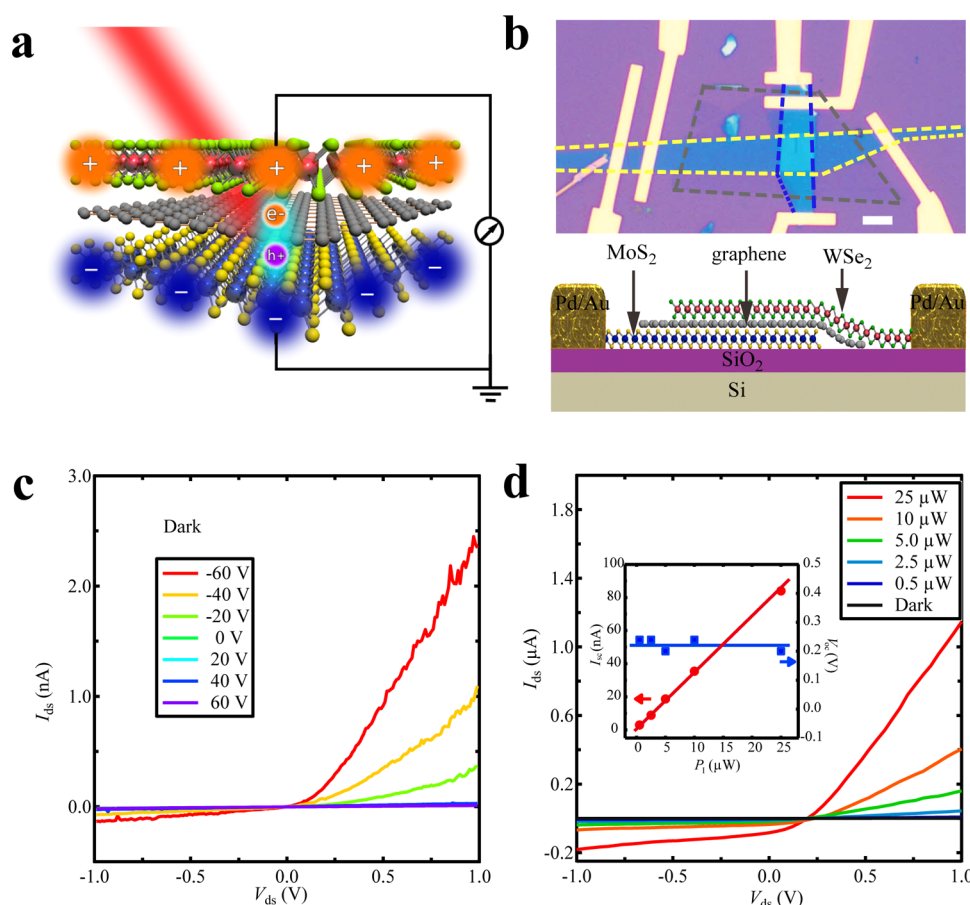


Figure 1. Broadband photovoltaic detector based on atomically thin p-g-n structure. (a) Schematic drawing of the p-g-n heterostructure for photodetection. (b) Top: optical image of a fabricated device based on van der Waals-assembled MoS₂-graphene-WSe₂ heterostructure. Scale bar, 5 μm . Bottom: schematic side view of the heterostructure. (c) $I_{\text{ds}}-V_{\text{ds}}$ characteristic curves without illumination, measured at various V_{g} (from -60 to 60 V, as denoted by different colors). (d) Output $I_{\text{ds}}-V_{\text{ds}}$ curves measured with exposure to a focused 488 nm laser beam (with power varying from 0.5 to 25 μW) at $V_{\text{g}} = 0$ V. Inset: extracted short-circuit current (I_{sc}) and open-circuit voltage (V_{oc}) versus incident power intensity.

sensitivity. Compared to previously proposed broadband solutions based on photoconductive graphene hybrid structures,²⁹ such a design could effectively suppress the dark current and overcome the issues of low specific detectivity and large energy consumption. As a demonstration, we fabricated an atomically thin MoS₂-graphene-WSe₂ heterostructure with a broadband photoresponse in the visible to short-wavelength infrared range at room temperature; this device exhibited a specific detectivity of up to 10^{11} Jones ($\text{cm Hz}^{1/2} \text{W}^{-1}$) in the near-infrared region, which meets the requirements of many important applications.

Figure 1b shows an optical image and schematic side view of the van der Waals-assembled MoS₂-graphene-WSe₂ heterostructure (see Methods for the fabrication details). The large-work-function metal Pd was used for doping electrodes to obtain p-doped WSe₂, whereas MoS₂ flakes remained n-type due to Fermi level pinning,³¹ as confirmed by their individual transfer curves (see Supporting Information section 3 for details). The current-voltage ($I_{\text{ds}}-V_{\text{ds}}$) characteristic curves of the heterojunction without light illumination were measured, and the results for back gate voltage V_{g} values from -60 to 60 V are plotted in Figure 1c. A rectifying behavior was observed. For this typical junction, I_{ds} increased with increasing V_{g} , indicating a p-type nature. In such device structure, the doping levels of the MoS₂ and WSe₂ flakes are tuned by the shared back gate, while separate tuning can be easily realized by

fabricating a transparent top gate. Upon exposure to a focused laser beam (488 nm), we observed a photovoltaic effect resulting in nonzero I_{ds} at zero bias, as shown in Figure 1d. The output $I_{\text{ds}}-V_{\text{ds}}$ curves were measured at $V_{\text{g}} = 0$ V with the incidence laser power varying from 0.5 to 25 μW . The short-circuit current (I_{sc}) was found to increase linearly with incident power intensity, whereas the open-circuit voltage (V_{oc}) (~ 0.23 V) was found to be independent of the laser power, as shown in the inset of Figure 1d. Considering that the voltage is applied across an atomically thin scale (~ 1 nm), the internal electric field is estimated to be extremely strong ($\sim 2 \times 10^8$ V m⁻¹). Thus, a p-n junction was formed within the vertically stacked MoS₂-graphene-WSe₂ heterostructure, as expected.

One of the most compelling features of such a p-g-n structure is the potential for realizing broadband photodetection by using graphene's gapless bandstructure. The main benefits of photodetectors include the photoresponsivity R and specific detectivity D^* . Here, D^* is defined by $(A\Delta f)^{1/2}R/i_{\text{n}}$, where A is the effective area, Δf is the electrical bandwidth, and i_{n} is the noise current. If assuming the noise from the dark current is a major contribution, D^* can be expressed as $R/(2qJ_{\text{d}})^{1/2}$, where J_{d} is the dark current density and q is the electron charge. We also confirmed such assumption is reasonable after we performed the measurement of noise current and obtained a comparable measured value (see details in Supporting Information section 4). Figure 2a

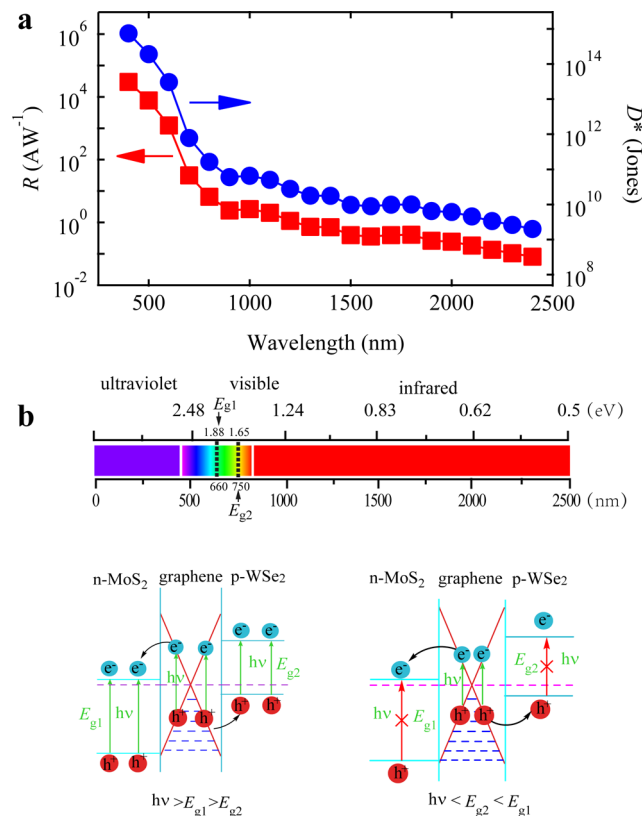


Figure 2. Broadband photoresponse. (a) Photoresponsivity R (left) and specific detectivity D^* (right) of a typical device for wavelengths ranging from 400 to 2400 nm. The device was tested in ambient air at $V_{ds} = 1$ V and $V_g = 0$ V. (b), Top: spectrum from ultraviolet to mid-infrared with the bandgap of monolayer MoS_2 (E_{g1}) and WSe_2 (E_{g2}) indicated. Bottom left: Schematic band diagrams and light absorption of the p–g–n heterostructure in the ultraviolet and visible range, in which the photon energy is larger than the band gaps of the TMD materials (E_{g1} and E_{g2}). Both the two TMDs and graphene respond to produce photogenerated free carriers. Bottom right: the case in which the wavelength is approaching the infrared region and the photon energy becomes smaller than E_{g2} . Only graphene responds to produce photogenerated free carriers.

presents the photoresponse results of a typical device in the visible to midwavelength infrared range at room temperature. Here, the laser wavelength was tuned continuously using a monochromator varying from 400 to 2400 nm with an interval of 100 nm. The device was tested in ambient air with $V_{ds} = 1$ V and $V_g = 0$ V. A remarkably strong response was acquired over the entire spectral range of interest, indicating that the p–g–n heterostructure can provide broadband photodetection, as designed. In the visible region, the measured photoresponsivity R was as high as 10^4 A W^{-1} , and the calculated specific detectivity D^* was as high as 10^{15} Jones. As the excitation laser wavelength increased, R and D^* decreased sharply, with R at approximately several A W^{-1} and D^* close to 10^{11} Jones in the near-infrared region. D^* continued to decrease slowly and became relatively stable at 10^9 Jones while approaching a wavelength of 2400 nm.

The wavelength dependence of the photoresponse can be explained by the limited spectral bandwidth of MoS_2 and WSe_2 used to build the atomically thin p–n junctions. Similar to conventional semiconductors, TMD materials exhibit no absorption¹⁷ when the photon energy is smaller than the bandgap. In the case of monolayer MoS_2 and WSe_2 , the band

gaps E_{g1} and E_{g2} are 1.88^{32,33} and 1.65 eV,^{34,35} corresponding to wavelengths of 660 and 750 nm, respectively. Figure 2b (left) schematically shows the light absorption of the p–g–n heterostructure in the visible range, in which the photon energy is larger than the bandgaps of the TMD materials. All three layered materials (MoS_2 , graphene, and WSe_2) are efficient absorption materials that produce abundant photo-generated free carriers, resulting in a considerably higher photoresponse. When the wavelength is near the infrared region and the photon energy is smaller than E_{g2} , the interband absorption of both monolayer MoS_2 and WSe_2 is forbidden. As shown in Figure 2b (right), the graphene layer becomes the only light absorption material to generate electron–hole pairs, yielding a relatively smaller photoresponse in the infrared range.

We further investigated the incident light power dependence of the photoresponse in both the visible and near-infrared regions, and the major results are plotted in Figure 3a and b, respectively (measured at $V_{ds} = 1$ V and $V_g = 0$ V in ambient conditions). Here, we introduce another key figure of merit: the external quantum efficiency (EQE), which is the ratio of the number of photoexcited charge carriers to the number of incident photons and can be expressed as $\text{EQE} = (hcI_p/e\lambda P_i)$, where h is the Planck constant, c is the speed of light, and λ is the wavelength of the incident laser. In the visible range (using a 532 nm laser), as shown in Figure 3a, both R (left axis) and the EQE (right axis) increase as the laser power P_i decreases and reach values of up to $4,250$ A W^{-1} and 1.0×10^6 %, respectively, at a power of 0.2 nW. For such weak incident light, D^* is as high as 2.2×10^{12} Jones, which is comparable to commercial InGaAs photodetectors ($\sim 10^{12}$ Jones with the requirement of cooling to 4.2 K for operation).^{36,37} In the near-infrared region (using a 940 nm laser), a similar trend was observed with relevant results shown in Figure 3b, in which R is as high as 306 mA W^{-1} at a power of 17 nW. These results imply possible practical applications of p–g–n heterostructures in multiple fields, as demonstrated by a customized room-temperature photoelectric imaging system obtained by replacing the CCD unit of a digital camera with our device and a connected preamplifier (Supporting Information section 7).

The structure of the ultrathin vertical p–g–n junction plays a crucial role in inducing an efficient photo gain mechanism and fast response over the entire spectrum. The photoexcited electron–hole pairs in the strong built-in electric field are spontaneously and rapidly separated.³⁸ The interlayer inelastic tunneling process occurs due to lateral momentum mismatch³⁹ in randomly stacked interfaces, which is beneficial for reducing interlayer recombination and increasing the carrier lifetime τ_{life} , thus achieving high gain G (for more details, see Supporting Information section 5). We also measured the response speed of the devices by varying the laser frequency from 1 to 1000 Hz (with results shown in Figure S5 in the Supporting Information). No obvious photocurrent decrease was observed while the laser pulse frequency was increased from 1 to 1000 Hz. We found the speed of photoresponse of our devices is very fast, with a rise time of 53.6 μs and a fall time of 30.3 μs , as shown in Figure 3c and d. In spite of this first demonstration of broadband photodetection based on atomically thin p–g–n junctions, further improvements in device performance can be reasonably expected upon optimization of the device parameters and further engineering of the material properties. For example, the device photoresponse could be enhanced by tuning both V_{ds} and V_g (see Supporting Information section 3

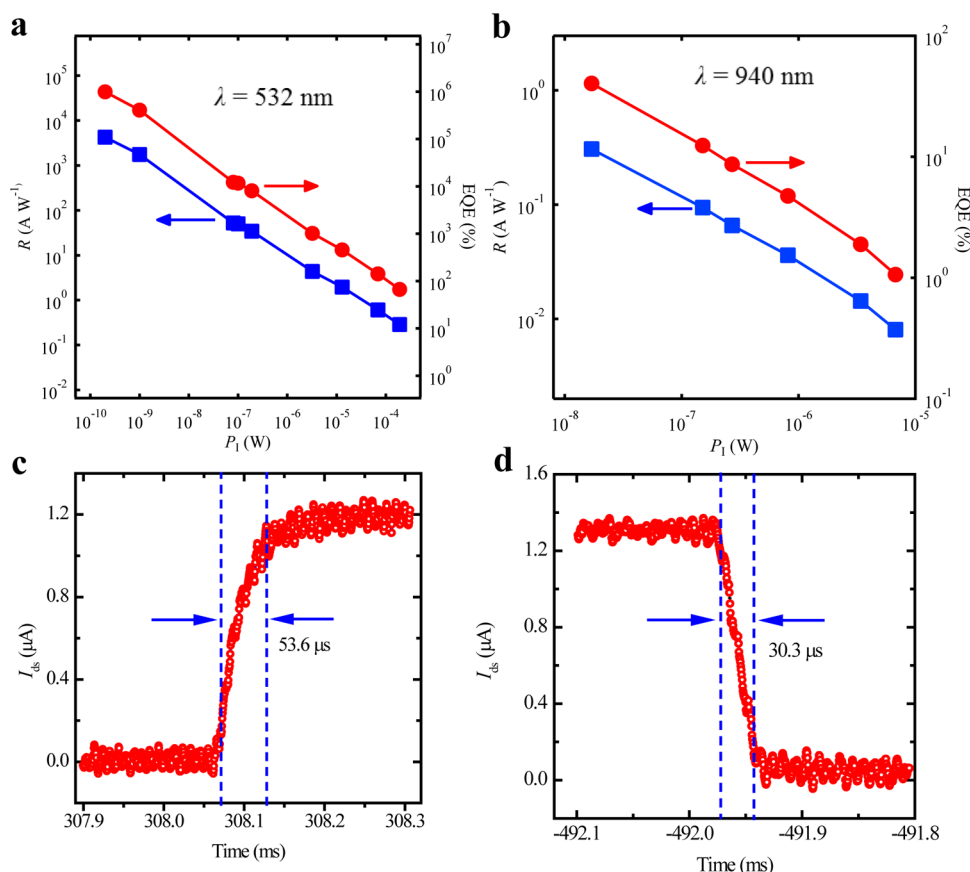


Figure 3. Illumination-power-dependent photoresponse. (a,b) Measured photoresponsivity R (left axis) and external quantum efficiency EQE (right axis) of a typical device versus illumination power P_1 in the visible region (a, using a 532 nm laser) and near-infrared region (b, using a 940 nm laser). Both R and EQE increase as P_1 decreases. The measurements were performed in ambient conditions with $V_{ds} = 1$ V and $V_g = 0$ V. (c, d) Fast photoresponse of a typical device measured by using a laser (637 nm) of 1000 Hz frequency. Here the rise/fall time was defined as the photocurrent increased/decreased from 10/90% to 90/10% of the stable photocurrent.

for details). The light absorption of graphene is highly sensitive to the Fermi level, which determines certain allowed interband excitations. Thus, improved device performance could be obtained by engineering the Fermi level to be located at the Dirac point.

To shed light on the photoresponse mechanism of the p–g–n photodetector, we conducted photocurrent mapping measurements by scanning the laser spot over a typical device. Figure 4a and b presents the photocurrent mapping at $V_{ds} = 0$ and 1 V, respectively, when $V_g = 0$ V. An optical micrograph of the device is shown in Figure 4c, which indicates the positions of the electrodes and different layered materials in the photocurrent images (highlighted by different colored lines). The photocurrent mapping measurements were performed under an 830 nm laser at a power of ~ 20.5 μ W under ambient conditions. The photocurrent mappings reveal that the strongest photoresponse occurs within the overlapped region (highlighted by red dashed lines) of MoS₂, graphene, and WSe₂, rather than the electrode regions. This suggests that the photoresponse in our devices originates from the designed atomically thin p–g–n junction, rather than the Schottky barriers formed at the metal contacts⁴⁰ or other device regions (see additional data sets in Supporting Information section 6).

In summary, we present a novel p–g–n heterostructure to enable broadband photovoltaic detection. Compared to previously proposed broadband solutions based on photoconductive structures, this design could effectively overcome

their limitations of low specific detectivity and large energy consumption. We successfully fabricated an atomically thin MoS₂-graphene-WSe₂ heterostructure and demonstrated broadband photoresponse in the visible to short-wavelength infrared range at room temperature along with competitive device performance. Other 2D materials other than graphene, such as black phosphorus, with features of smaller band gap (than TMD materials) and being polarization-sensitive²⁷ also could be used as the sandwiched light absorption material in such atomic thin p–n junctions. Our results underscore the great potential of van der Waals junctions based on 2D materials for future important optoelectronic applications.

METHODS

We used a standard mechanical exfoliation method to isolate mono/few-layer MoS₂, WSe₂, and graphene films. The thickness of the flakes was first measured using a Bruker Multimode 8 atomic force microscope (AFM) and further confirmed by Raman and photoluminescence measurements (Supporting Information section 2). The vertical atomically thin heterostructures were obtained using a polymer-free van der Waals assembly technique followed by a transfer process to a Si wafer covered by a 300 nm thick SiO₂ layer (Supporting Information section 1). A conventional electron-beam lithography process (FEI F50 with Raith pattern generation system) followed by standard electron-beam evaporation of metal

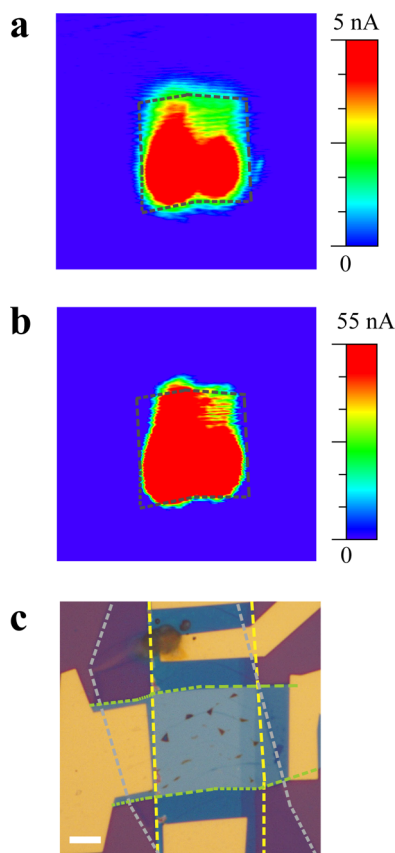


Figure 4. Photocurrent mapping of the p–g–n photodetector. (a, b) Photocurrent mapping results for a typical device at $V_{ds} = 0$ V (a) and $V_{ds} = 1$ V (b) with $V_g = 0$ V. Measurements were performed under an 830 nm laser at a power of ~ 20.5 μ W in ambient conditions. The photocurrent mappings show the strongest photoresponse within the overlapped p–g–n region (highlighted by gray dashed lines). (c) Optical microscopy image of the device with the measurements performed. MoS₂, graphene, and WSe₂ are highlighted by yellow, light gray, and green dashed lines, respectively. Scale bar is 5 μ m.

electrodes (typically 30 nm Pd/30 nm Au) was used to fabricate the vertical p–g–n heterostructure devices.

Electrical transport measurements were performed using a Keithley 2636A dual channel digital source meter. Photoresponse measurements were performed under ambient conditions using a Lake Shore probe station. The wavelength-dependent photoresponse in Figure 2a was measured using a supercontinuum full spectrum white light laser source (400–2400 nm) combined with a monochromator. The spot size was ~ 50 μ m.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.5b04538.

Experimental details about van der Waals heterostructure fabrication; material characterization (Raman, AFM, and PL spectra); individual MoS₂ and WSe₂ transfer curves; additional photoresponsivity measurements as a function of wavelength, bias and back gate voltage; time-resolution photovoltaic response; additional infrared photocurrent mapping; room-temperature photoelectric imaging (PDF)

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Author Contributions

M. L., F. M., and W. H. conceived the project and designed the experiments. M. L., E. L., P. W., A. G., W. L., and J. Z. performed device fabrication and characterization. P. W. and W. H. performed microscopic photocurrent mapping and room-temperature photoelectric imaging. M. L., W. H., F. M., and B. W. performed data analysis and interpretation. Y. L. and S. Y. performed the WSe₂ single crystal growth. F. M., W. H., M. L., and B. W. cowrote the paper, and all authors contributed to the discussion and preparation of the manuscript.

Notes

The authors declare no competing financial interest.

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